# Cover letter

Dear Editor,

Please consider our original research manuscript, “TSHR and IGF1R in Graves disease and thyroid eye disease: genetic evidence from blood expression”.

Prior genetic studies have already implicated TSHR in Graves disease and ophthalmopathy. Our study examines how consistently blood expression-related evidence supports TSHR and IGF1R when evaluated within the same analysis framework across distinct disease definitions. We screened 2,544 potentially druggable genes and compared five disease datasets, explicitly acknowledging overlapping FinnGen participants. TSHR associations were consistent in direction, whereas IGF1R evidence remained limited. CTLA4, HLA-DQA2 and TUBB also passed the full-screen threshold in both GD datasets. Instrument and allele-frequency analyses qualify the robustness of these findings; secondary colocalization limits claims of expression-mediated causality. The contribution is a focused comparison and sensitivity assessment using public resources, not a claim to have discovered TSHR susceptibility or established a new orbital mechanism.

The manuscript distinguishes inherited expression-related susceptibility from treatment efficacy and does not claim a TED-specific effect within GD. External TSHR A-subunit and orbital fibroblast experiments provide biological context, while the small in-house orbital tissue series remains descriptive. Three main tables, three main figures, a supplementary document with three tables and two figures, and one detailed results workbook accompany the manuscript.

This version is prepared for author review. Before submission, the corresponding author will confirm the target journal, all-author approval, originality and exclusive-submission declarations, current funding and conflict disclosures, and the status and extent of overlap with any related orbital-transcriptome manuscript. These declarations should not be inferred from completion of the computational analysis.

Sincerely,

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